

Asking “Which Is Worse?” Rarely Reverses Four Sub-2B Qwen Judges: A Letter-Calibrated Negation Test on a Code-Heavy RewardBench Subsample

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Abstract—Small open language models are attractive as cheap pairwise judges, but does their verdict depend on the criterion asked? We test this by negation: each pair is judged under “Which response is better?” and “Which response is worse?”, in both orders. We evaluate four Qwen judges (Qwen2.5-0.5B/1.5B-Instruct, Qwen3-0.6B/1.7B; thinking off; A-vs-B logit readout) on 300 RewardBench pairs, 217 of which are code pairs whose responses differ by a median of one word. Under argmax, a correct reversal occurs in 0, 0, 0 and 1 of 300 pairs, and the pre-registered “position-locking” finding is largely implied by letter rates. A post-hoc order-averaged logit readout (existing swap calibration) shows that the pair-dependent shift under “worse” is positively correlated with the shift under “better” (Pearson $r = 0.95, 0.63, 0.90, 0.63$). For the three smaller judges (calibrated accuracy 0.50–0.61) the shift keeps most of its size; Qwen3-1.7B (accuracy 0.81) keeps its sign in 0.76 of pairs but the shift shrinks to about a third. A larger Qwen3-4B judge, run on one 100-pair resample as a positive control, behaves differently: $r = 0.10$ [−0.09, 0.30], same sign 0.59, calibrated accuracy 0.86, calibrated reversal 0.31 [0.23, 0.40]; on the same 100 pairs its r is 0.475 lower than Qwen3-1.7B’s (paired bootstrap 95% CI [0.229, 0.720]). It shows no negative correlation, so it is only a partial positive control, and with one family, one resample and $n = 100$ the size trend is suggestive, not established. A greedy-generation check for Qwen3-1.7B agrees with the logit argmax on 90% of 400 judgments.

Index Terms—LLM-as-a-judge, pairwise evaluation, criterion negation, position bias, calibration, small language models, negative result

I. INTRODUCTION

Pairwise LLM judges are widely used to compare model outputs [1], and their position bias is well documented [2], [3], [4]. Small open models (below 2B parameters) are attractive judges because they run on commodity CPUs, and recent work evaluates Qwen3 models as small as 0.6B as judges on RewardBench [5]. A basic precondition for any judge is that its verdict depends on the criterion it is asked about. Accuracy on a “which is better?” prompt cannot establish this, and an order-swap test detects position bias but not whether the judge reads the question.

We probe criterion conditioning by negation, a property known as negation invariance in the logical-consistency literature [6]: a judge that prefers the chosen response under “better” should prefer the rejected one under “worse”. Asking both

questions in both orders gives four judgments per pair. Our first analysis, which was pre-registered, classified each pair by the four argmax verdicts and concluded that the smallest judges are “position-locked”. Referee feedback showed that this conclusion was driven by the readout: strongly letter-biased judges produce the same letter four times regardless of what they prefer. This revision therefore separates the letter prior from the pair-dependent part of the logit by averaging over presentation orders, and finds a pair-dependent shift that mostly keeps its sign under both questions.

Contributions.

- 1) A four-judgment negation design with two readouts: a pair-level argmax taxonomy and an order-averaged logit difference that removes a constant letter offset; the latter applies existing swap calibration [2], [7], [8] and is not a new method (Section III).
- 2) An evaluation of four sub-2B Qwen judges (thinking off) on 300 RewardBench pairs with pinned model and data revisions (Section IV).
- 3) The finding, for these judges and this sample, that the order-averaged logit shift under “worse” is positively correlated with that under “better”; for the three smaller judges it also keeps most of its magnitude, while Qwen3-1.7B shows an attenuated shift (either weak partial reversal or lower confidence under “worse”). The correlation persists on non-code pairs and high-mass pairs (a self-selected subset for Qwen3-1.7B) (Section V).
- 4) A correction of our own pre-registered framing: under a letter-biased null the position-locking hypothesis is nearly implied by letter marginals, so argmax taxonomies can misattribute a criterion-insensitive pair-dependent shift to position bias (Sections V and VI).
- 5) A larger-judge check (Qwen3-4B, 100 pairs, one resample) showing that the calibrated readout registers a much weaker positive coupling for this judge than for Qwen3-1.7B on the same pairs, so the test is not blind to between-judge differences; the 4B judge is not a clean criterion follower (Section V-A).

II. RELATED WORK

Negation and logical consistency of judges. Liu et al. [6] propose measuring logical preference consistency of LLM judgments by transitivity, commutativity and negation invariance (whether a model inverts its judgment when the relation is negated), evaluate open models from Phi-3-mini to 13B and two API models, and introduce the REPAIR data augmentation. Our “better” vs. “worse” test is an instance of negation invariance; we add order swaps, sub-2B judges and a letter-calibrated readout. GRP [9] asks strong closed judges which answer is *worse* as a prompting method and reports improved consistency on JudgeBench. Bagaria et al. [10] reverse the rubric criterion in pointwise judging with Qwen2.5-7B-Instruct and report desired verdict flips of 16.8% and 32.2% (two candidate models) under criterion reversal. PRIN [11] contrasts “which are correct?” with “which are incorrect?” in multiple-choice settings with models of 8B parameters and larger. Elkins and Chun [12] audit negated vs. affirmative framings of moral dilemmas in 16 models and report swings of up to 76 percentage points for small (1–4B) models; their task is stance-taking, not response comparison. JudgeSense [13] measures judge sensitivity to prompt rewording more broadly.

Position bias and hidden signal. Zheng et al. [1] report position and verbosity bias; Wang et al. [2] propose balanced-position calibration, which our order-averaged readout resembles; Shi et al. [3] study position bias across 15 judges; Norman et al. [4] evaluate 21 judges from nine providers and report high test-retest reliability coexisting with severe position bias in two production-deployed judges. Mukherjee and Sitaram [14] study 64 open-weight judges and report that on LLMBar [15] the verdicts of 50 judges agree with human labels only 0.456 of the time even after averaging both orders, while probes on activations recover much more signal. Our calibrated analysis makes a related, simpler observation at the output layer: the argmax verdict of a letter-biased judge hides a content preference that the logit difference still carries.

Small judges and readouts. SLMJury [16] benchmarks 16 small judges (0.6B–14B, four families) on ten benchmarks; in the parts we read, verdicts are parsed from generated text, and we found no logit scoring, order swap or criterion reversal. Xie et al. [17] compare generative verdicts, Yes/No log-probability margins and linear probes for Qwen3 judges from 0.6B to 8B in pointwise rubric judging; for Qwen3-1.7B on RaR-Science-Static they report criterion-level macro-F1 of 0.443 (generative), 0.449 (logprob) and 0.835 (probe). At that size and on that task, the logprob readout is thus close to the generative one, which is weak, indirect support for our logit readout but not a substitute for validating it on our task.

Swap calibration. Balanced-position calibration [2] evaluates both orders, and LLM-RankFusion [7] queries both permutations of a pair and combines the “A” and “B” token probabilities into an order-agnostic preference. Our d_q is an application of this precedent to the A-minus-B logit, used here as a diagnostic, not proposed as a new method. Wang et al. [8] show that taking the mean of the judgment distribution

beats the greedy mode across pointwise, pairwise and listwise settings; our order-averaged logit is in the same spirit (a distribution-level readout rather than argmax), but we use it to probe criterion sensitivity, not to improve accuracy.

Negation in language models. Pretrained LMs often fail to distinguish negated from affirmative probes [18], and larger LMs can perform worse on negated prompts [19].

Difference from prior work. Negation invariance of judges [6], reversed-goal prompting [9] and swap calibration [2], [7] are established ideas, so our contribution is narrow: applying the negation test in both orders to four judges below 2B parameters with a logit readout, and showing that order-averaging changes the diagnosis from position-locking to a pair-dependent shift that the negated question does not invert. Our literature scan used arXiv and web search only; we do not claim to be first.

III. METHOD

Prompt. Each judgment uses the model’s chat template with a single user turn containing the user request, “[Response A]” and “[Response B]”, a question, and “Answer with a single letter: A or B.” The questions are *better* (“Which response is better?”), *W1* (“Which response is worse?”, primary negation), *W2* (“Which response should be rejected?”) and *length* (“Which response is longer?”, control). For Qwen3 we pass `enable_thinking=False`.

Readout. One forward pass; the score s is the difference of next-token logits of the single tokens “A” and “B”; the argmax verdict is the letter with the higher logit. The probability mass on {“A”, “A”, “B”, “B”} is recorded as a format-compliance measure. Inference is deterministic.

Four judgments per pair. For each pair (chosen C , rejected R) we ask *better* and *W1*, each with C shown as A (order *cf*) and with R shown as A (order *rf*).

Argmax taxonomy (pre-registered). Each pair receives the first matching category: (1) *correct reversal*: C in both orders under *better* and R in both orders under *W1*; (2) *position-locked*: the same letter in all four judgments; (3) *criterion-blind*: the same response in all four; (4) *reversed-consistent*; (5) *other*. Under a uniformly random judge the shares are 1/16, 2/16, 2/16, 1/16 and 10/16. *Position-consistent accuracy* is the share of pairs with C chosen in both orders under *better* (the conditioning subset, size n_{cond}); the *conditional flip rate* is the share of that subset for which *W1* selects R in both orders.

Letter-calibrated readout (post hoc). For each pair and question q , $d_q = (s_q^{cf} - s_q^{rf})/2$. If the judge adds a constant offset to the “A” logit, it cancels; $d_q > 0$ means the judge favors C as the answer to q . A criterion-following judge has $d_{\text{better}} > 0$ and $d_{\text{W1}} < 0$ on most pairs, i.e. a negative correlation; a criterion-blind judge, which favors the same response whatever the question, has a positive correlation. We report Pearson $r(d_{\text{better}}, d_{\text{W1}})$, the same-sign share, calibrated accuracy (share with $d_{\text{better}} > 0$) and calibrated reversal (share with $d_{\text{better}} > 0$ and $d_{\text{W1}} < 0$), with 95% bootstrap CIs over pairs (2000 resamples, seed 0).

TABLE I
JUDGES AND DATA WITH PINNED REVISIONS (FIRST 12 HEX DIGITS; FULL SHAS IN RESULTS/MODELS.MD).

Repository	Revision	Params	License
Qwen/Qwen2.5-0.5B-Instruct [20]	7ae557604adf	494,032,768	apache-2.0
Qwen/Qwen2.5-1.5B-Instruct [20]	989aa7980e4c	1,543,714,304	apache-2.0
Qwen/Qwen3-0.6B [21]	c1899de289a0	751,632,384	apache-2.0
Qwen/Qwen3-1.7B [21]	70d244cc86cc	2,031,739,904	apache-2.0
Qwen/Qwen3-4B [21] (r1 only)	1cfa9a720891	not recorded	apache-2.0
allenai/reward-bench (data) [22]	168d848cddbbe	–	odc-by

Magnitude and order-averaged probabilities (post hoc, referee round 2). The correlation ignores scale, so we also report the OLS slope of d_{W1} on d_{better} (with intercept; criterion-following $\Rightarrow$ negative, unchanged shift $\Rightarrow$ 1), the median over pairs of $|d_{W1}|/|d_{\text{better}}|$, and order-averaged probabilities $P_C(q) = [\sigma(s_q^{cf}) + \sigma(-s_q^{rf})]/2$ of picking C (two-way A-vs-B softmax). For $S = P_C(\text{better}) + P_C(W1)$, a criterion-following judge gives $S \approx 1$ with $P_C(\text{better})$ far from 0.5, and a criterion-blind judge gives $S \approx 2P_C(\text{better})$; but a judge that follows only the letter gives $P_C = 0.5$ under both questions and also $S = 1$, so $S \approx 1$ alone does not discriminate. We therefore also report $\Delta = P_C(\text{better}) - P_C(W1)$ (blind ≈ 0) and $|P_C(\text{better}) - 0.5|$. Within-family size differences (larger minus smaller judge) use a paired bootstrap over the same 300 pairs. For HumanEvalPack pairs we compute the token-level Levenshtein distance between C and R (whitespace tokens).

Letter-biased null. For each judge, an independent-letter judge with the observed per-question “A” rates p_b (better) and p_w (W1) gives all four judgments the same letter with probability $p_b^2 p_w^2 + (1 - p_b)^2 (1 - p_w)^2$ and has position-consistent accuracy $p_b(1 - p_b)$.

Hypotheses (pre-registered). H1: the conditional flip rate rises with size within each family (paired bootstrap, Bonferroni-adjusted 97.5% CIs); declared inconclusive if any $n_{\text{cond}} < 30$. H2: among non-correct-reversal pairs, the position-locked share exceeds 0.5 (95% CI lower bound > 0.5), for Qwen2.5-0.5B and Qwen3-0.6B. The calibrated readout, letter-biased null, magnitude, size-difference, length and edit-distance analyses were added after review and are not pre-registered. A development-set A/B-mass exclusion check (< 0.5) was run for Qwen2.5-0.5B only (20 pairs, mean mass 0.9998), a protocol deviation; on the evaluation set all judges have mean mass ≥ 0.745 .

IV. EXPERIMENTAL SETUP

Models. Four judges loaded from safetensors with `trust_remote_code=False` in float32 (Table I); all Apache-2.0 per their model cards. A fifth judge, Qwen3-4B, was added later as an intended positive control and run on resample 1 only (100 pairs $\times$ 4 judgments, same prompt, readout and `enable_thinking=False`).

Data. RewardBench [22] (ODC-By) at revision 168d848cddbba9764fae4a544dc9calle6cca4931, sections chat, chat-hard and reasoning. Of 2985 items we

excluded 740 in safety subsets and 1 matching a safety keyword list, 1021 with a response longer than 150 tokens, and 213 whose full prompt exceeded 400 tokens, leaving 1010 (chat 46, chat-hard 275, reasoning 689). We drew 30 development pairs (format checks only) and three disjoint resamples of 100 evaluation pairs (sampling seeds 0, 1, 2), pooled to 300: chat 12, chat-hard 65, reasoning 223. **Composition caveat:** the length filter makes the sample code-heavy: 217 of 300 pairs are HumanEvalPack pairs (correct vs. buggy code for the same task), and 83 are other pairs. The HEP responses are near-identical: the token edit distance between C and R has 10/25/50/75/90% quantiles 1/1/1/2/3 (maximum 24), 131 of 217 differ by exactly one token and 204 by fewer than five; the non-HEP pairs have a median distance of 45. HEP results are therefore about one-word bug edits. We report results on both subsets.

Passes. Per model, 300 pairs $\times$ 4 judgments plus, on resample 0 only, 100 pairs $\times$ 2 orders for W2 and for the length control: 1600 passes per model. Inference is deterministic, so the resamples are data splits, not seeds; there is one pass per prompt and no seed variance. Uncertainty is quantified by bootstrap over pairs. The calibrated analysis reuses the same raw outputs; no new model runs.

Hardware and budget. CPU only: one shared aarch64 machine, 4 threads, float32; Python 3.14.4, torch 2.14.1, transformers 4.57.6 (results/pip_freeze.txt). Summed per-pass forward time is 2.37 h (879 s, 3281 s, 1153 s, 3222 s). Extension judges (Qwen2.5-3B, Llama-3.2-1B) were allowed only if the core sweep was projected at ≤ 1.5 h; the projection was 2.32 h, so none was run. The later Qwen3-4B run (400 passes) took 1848 s of summed forward time on the same machine.

V. RESULTS

Argmax readout (pre-registered). Table II gives the pooled taxonomy. Correct reversal is 0.000, 0.000, 0.000 and 0.003 (1 of 300 pairs, Qwen3-1.7B), below the random-judge 0.062. Position-consistent accuracy under *better* is 0.000 to 0.430. H2 is supported for both pre-specified judges: the position-locked share of non-correct-reversal pairs is 1.000 [1.000, 1.000] for Qwen2.5-0.5B, which answers “A” on all 1600 passes (its row equals the always-A reference), and 0.890 [0.857, 0.927] for Qwen3-0.6B (exploratory: Qwen2.5-1.5B 0.710 [0.657, 0.763], Qwen3-1.7B 0.431 [0.378, 0.487]). H1 is inconclusive by the pre-registered rule ($n_{\text{cond}} = 0$ and 15 for the small judges); the Qwen3 conditional flip rate moves from 0.000 to 0.008 (97.5% CI of the difference [0.000, 0.030]), and the chance-normalized unconditional flip differences have 97.5% CIs [0.000, 0.040] (Qwen2.5) and [−0.018, 0.053] (Qwen3). Under the argmax readout the flip rates are therefore near zero at both sizes; the calibrated readout does show a size difference (below), so the two readouts disagree on this point.

H2 is close to tautological. The letter rates explain most of the argmax picture. Per-question “A” rates (better/W1) are 1.00/1.00, 0.20/0.01, 0.94/0.94 and 0.34/0.12 (Fig. 1). An independent-letter judge with these rates gives all four

TABLE II

ARGMAX READOUT, POOLED OVER 300 REWARD BENCH PAIRS PER JUDGE (W1 = “WHICH RESPONSE IS WORSE?”). SHARES OVER ALL 300 PAIRS. n_{cond} : PAIRS CORRECT UNDER “BETTER” IN BOTH ORDERS. REFERENCE ROWS ARE ANALYTIC. “UNDEF.”: $n_{\text{cond}} = 0$. POSITION-LOCKED SHARES ARE LARGELY PREDICTED BY LETTER RATES (TABLE III); SEE TEXT.

Judge	n_{cond}	Acc. (better)	Cond. flip	Uncond. flip	Correct rev.	Pos.-locked	Crit.-blind	Rev.-consist.	Other	A/B mass
Qwen2.5-0.5B	0	0.000	undef.	0.000	0.000	1.000	0.000	0.000	0.000	1.000
Qwen2.5-1.5B	31	0.103	0.000	0.013	0.000	0.710	0.003	0.000	0.287	0.996
Qwen3-0.6B	15	0.050	0.000	0.027	0.000	0.890	0.023	0.000	0.087	0.975
Qwen3-1.7B	129	0.430	0.008	0.040	0.003	0.430	0.113	0.000	0.453	0.745
Random judge	–	0.250	0.250	0.250	0.062	0.125	0.125	0.062	0.625	–
Always-A judge	0	0.000	undef.	0.000	0.000	1.000	0.000	0.000	0.000	–

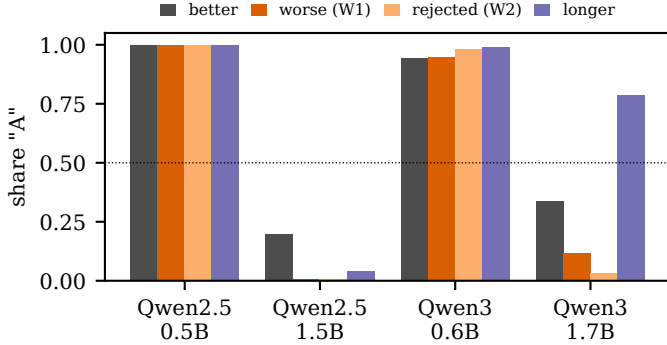


Fig. 1. Share of passes answered “A” per question (better and W1: 600 passes per judge; W2 and length: 200 passes, resample 0 only). Dotted line: 0.5.

judgments the same letter in 1.00, 0.63, 0.79 and 0.34 of pairs; observed: 1.00, 0.71, 0.89 and 0.43 (Table III). Against the uniform random judge (0.125) H2 is therefore nearly guaranteed for any strongly letter-biased judge, and its support tells us about letter bias, not about the criterion. Likewise, position-consistent accuracy under the letter-biased null is 0.000, 0.159, 0.053 and 0.224, close to the observed 0.000, 0.103 and 0.050 for the three smaller judges; only Qwen3-1.7B (0.430) exceeds what its letter rates predict.

Calibrated readout: the shift is not inverted. Table III and Fig. 2 show the letter-calibrated analysis. For all four judges, d_{W1} is positively correlated with d_{better} : $r = 0.95$ [0.92, 0.97], 0.63 [0.51, 0.73], 0.90 [0.87, 0.93] and 0.63 [0.55, 0.71]. The same sign occurs in 0.92, 0.71, 0.87 and 0.76 of pairs, and every CI lower bound is above 0.5 (0.88, 0.66, 0.82, 0.71). A criterion-following judge would show a negative correlation. Thus, once the letter offset is removed, each pair still produces a shift d whose sign is mostly unchanged by the question; the question mainly moves the letter prior. This shift need not be a quality judgment: calibrated “better” accuracy is 0.61, 0.50, 0.57 and 0.81, i.e. near chance for the three smaller judges, and only Qwen3-1.7B shows clear content signal under *better*, which it does not invert under W1. For Qwen2.5-0.5B, d is also tiny compared with the letter offset: the median $|d_{\text{better}}|$ is 0.047 logits against a mean order-averaged A-minus-B score of 2.80 (Qwen3-1.7B: median $|d_{\text{better}}|$ 5.4). Its $r = 0.95$ therefore shows that small residual shifts are reproduced across questions, not that this

judge has a coherent preference between the responses. The calibrated-reversal shares (0.06 to 0.22) are the tail where d changes sign; we defined no null for them and do not interpret them as evidence of reversal. Under the argmax taxonomy the three smaller judges look position-locked rather than criterion-blind (criterion-blind shares 0.000 to 0.113 for all four judges in Table II) because a large letter offset dominates the argmax verdict.

Magnitude (post hoc). Correlation alone does not say whether the shift keeps its size. Table IV gives the OLS slope of d_{W1} on d_{better} : 0.82, 0.39, 0.67 and 0.26, all with CIs above 0 (a criterion-following judge would have a negative slope). For the three smaller judges the median $|d_{W1}|/|d_{\text{better}}|$ is 0.82–0.90, so the pair-dependent shift is largely kept under “worse”; their order-averaged probabilities stay close to 0.5 (mean $|P_C(\text{better}) - 0.5| \leq 0.083$), and Δ is 0.000 to 0.013. For Qwen3-1.7B the shift shrinks to about a third under “worse” (median ratio 0.35, slope 0.26) without usually changing sign: $P_C(\text{better}) = 0.678$ and $P_C(W1) = 0.543$, so $S = 1.22$ [1.18, 1.26] and $\Delta = 0.135$ [0.11, 0.16]. A full reversal would put $P_C(W1)$ near $1 - P_C(\text{better})$; the observed value lies between the blind and the following case, closer to the blind one. We read this as an attenuated shift, consistent with a weak partial reversal but also with lower confidence under “worse”. The $S \approx 1$ values of the three smaller judges (1.002, 0.997, 1.014) are not evidence of criterion following, because a letter-only judge also gives $S = 1$; their $\Delta \approx 0$ and $P_C \approx 0.5$ fit the letter-dominated reading.

Size within family (post hoc). Unlike the argmax H1 analysis, the calibrated readout shows a size difference: r and the same-sign share fall with size in both families. Paired bootstrap differences (larger minus smaller judge, 95% CIs) are, for r , -0.32 [$-0.43, -0.22$] (Qwen2.5) and -0.27 [$-0.37, -0.18$] (Qwen3), and for the same-sign share -0.21 [$-0.27, -0.15$] and -0.10 [$-0.16, -0.05$]. Calibrated “better” accuracy moves in opposite directions: -0.12 [$-0.19, -0.04$] for Qwen2.5 and $+0.23$ [0.17, 0.29] for Qwen3. This trend is post hoc, rests on two sizes in each of two families, and is confounded with large differences in letter bias between the judges (Fig. 1); it is not evidence of a general scaling law.

Robustness. (i) *Sample composition*: the correlation is 0.57–0.95 on the 83 non-HEP pairs and 0.58–0.96 on the 217 HEP pairs (Table III). Calibrated “better” accuracy is lower

TABLE III

LETTER-CALIBRATED ANALYSIS (POST HOC), $d_q = (s_q^{cf} - s_q^{rf})/2$. r : PEARSON CORRELATION OF d_{better} AND d_{W1} (CRITERION-FOLLOWING $\Rightarrow r < 0$; CRITERION-BLIND $\Rightarrow r > 0$). SAME SIGN: SHARE WITH $\text{sign}(d_{\text{better}}) = \text{sign}(d_{W1})$. CAL. ACC.: SHARE WITH $d_{\text{better}} > 0$. CAL. REV.: SHARE WITH $d_{\text{better}} > 0$ AND $d_{W1} < 0$ (NO NULL DEFINED). HEP: HUMANEVALPACK. MASS>0.9: PAIRS WITH A/B MASS > 0.9 ON ALL FOUR PASSES. LETTER NULL: SHARE OF PAIRS WITH THE SAME LETTER IN ALL FOUR JUDGMENTS, EXPECTED UNDER INDEPENDENT LETTERS AT OBSERVED RATES / OBSERVED. 95% BOOTSTRAP CIs IN BRACKETS.

Judge	All pairs ($n = 300$)				Non-HEP ($n = 83$)	HEP ($n = 217$)	Mass>0.9	Letter null
	r	Same sign	Cal. acc.	Cal. rev.	r	r	$n; r$	exp. / obs.
Qwen2.5-0.5B	0.95 [0.92, 0.97]	0.92 [0.88, 0.95]	0.61	0.06 [0.03, 0.08]	0.95 [0.92, 0.97]	0.96 [0.94, 0.97]	300; 0.95 [0.92, 0.97]	1.00 / 1.00
Qwen2.5-1.5B	0.63 [0.51, 0.73]	0.71 [0.66, 0.76]	0.50	0.22 [0.18, 0.27]	0.72 [0.61, 0.82]	0.58 [0.38, 0.72]	288; 0.60 [0.47, 0.71]	0.63 / 0.71
Qwen3-0.6B	0.90 [0.87, 0.93]	0.87 [0.82, 0.90]	0.57	0.08 [0.05, 0.11]	0.91 [0.88, 0.94]	0.91 [0.84, 0.95]	286; 0.91 [0.88, 0.94]	0.79 / 0.89
Qwen3-1.7B	0.63 [0.55, 0.71]	0.76 [0.71, 0.81]	0.81	0.18 [0.14, 0.23]	0.57 [0.38, 0.72]	0.69 [0.59, 0.77]	140; 0.63 [0.50, 0.73]	0.34 / 0.43

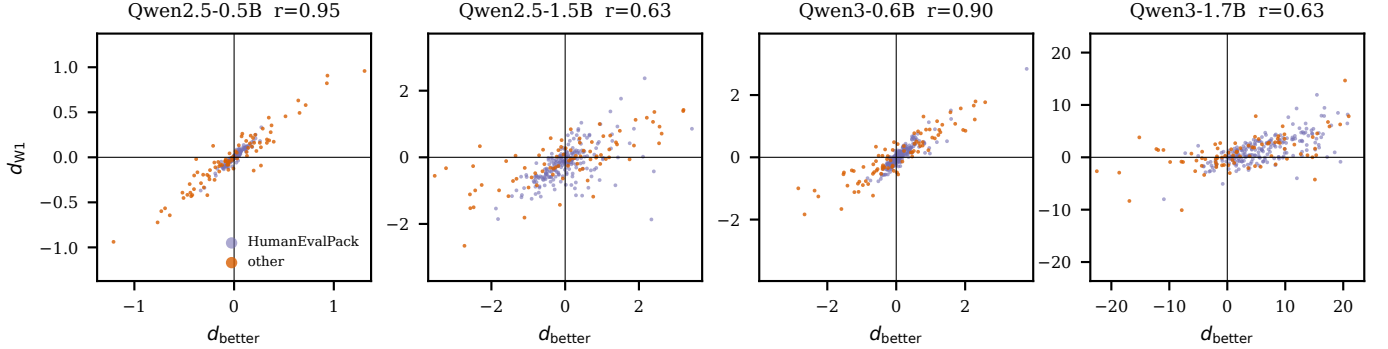


Fig. 2. Letter-calibrated preference for the chosen response under “better” (x) vs. “worse” (y), one point per pair ($n = 300$ per judge; purple: HumanEvalPack code pairs, orange: other). A judge that follows the criterion would place points in the lower-right quadrant; all four judges show a positive slope.

TABLE IV

MAGNITUDE OF THE CALIBRATED SHIFT (POST HOC, $n = 300$ PER JUDGE). SLOPE: OLS OF d_{W1} ON d_{better} . RATIO: MEDIAN $|d_{W1}|/|d_{\text{better}}|$. P_C : ORDER-AVERAGED PROBABILITY OF PICKING C . $S = P_C(\text{better}) + P_C(W1)$; $\Delta = P_C(\text{better}) - P_C(W1)$. 95% BOOTSTRAP CIs.

Judge	Slope	Ratio	P_C better / W1	S	Δ
Qwen2.5-0.5B	0.82 [0.77, 0.87]	0.90	0.501 / 0.501	1.002 [1.00, 1.01]	-0.000 [-0.00, 0.00]
Qwen2.5-1.5B	0.39 [0.31, 0.48]	0.82	0.505 / 0.492	0.997 [0.98, 1.01]	0.013 [0.00, 0.02]
Qwen3-0.6B	0.67 [0.61, 0.73]	0.89	0.511 / 0.504	1.014 [1.00, 1.03]	0.007 [0.00, 0.01]
Qwen3-1.7B	0.26 [0.20, 0.31]	0.35	0.678 / 0.543	1.220 [1.18, 1.26]	0.135 [0.11, 0.16]

on non-HEP pairs (0.43, 0.53, 0.48, 0.61) than on HEP pairs (0.68, 0.48, 0.61, 0.88), so the content signal of Qwen3-1.7B is stronger on code pairs; the positive correlation is present in both subsets. Within HEP, by edit distance, r is 0.96, 0.70, 0.88, 0.70 at distance 1 ($n = 131$), 0.95, 0.26, 0.90, 0.68 at distance 2 ($n = 50$) and 0.97, 0.63, 0.95, 0.67 at distance ≥ 3 ($n = 36$); Qwen3-1.7B’s calibrated accuracy is 0.89, 0.96 and 0.75 in these bins. Only 13 HEP pairs differ by five or more tokens (r 0.97, 0.18, 0.99, 0.40, with wide CIs), so we cannot say how the judges behave on HEP pairs with larger edits. (ii) *Format compliance*: restricting to pairs with A/B mass > 0.9 on all four passes ($n = 300, 288, 286, 140$) gives $r = 0.95, 0.60, 0.91, 0.63$. For Qwen3-1.7B, whose mean mass is 0.745 and 25.2% of whose better/W1 passes have mass below 0.5, this filter keeps 140 of 300 pairs and selects pairs with lower calibrated accuracy (0.72 vs. 0.81 overall), so the subset is not representative. (iii) *Partly checked*: the logit readout was compared with greedy generated verdicts for Qwen3-1.7B only

(90% judgment agreement, pair-category agreement 0.77); not for the other three judges.

Length control as a partial positive control (resample 0, exploratory). Under the argmax readout, on 72 pairs with different character lengths, the longer response is picked in both orders in 0.000, 0.000, 0.014 and 0.250 of pairs (random-judge level 0.25). With the calibrated readout, d_{length} (“Which response is longer?”) correlates with the true character-length difference at $r = 0.12 [-0.20, 0.40]$, 0.68 [0.46, 0.81], 0.60 [0.33, 0.76] and 0.46 [0.20, 0.64] ($n = 72$), and has the sign of the longer response in 0.65, 0.61, 0.64 and 0.58 of pairs. So three judges’ calibrated shift carries some length information. It is not a clean positive control: d_{length} tracks d_{better} closely ($r = 0.96, 0.89, 0.91, 0.74$ on the 100 pairs), and d_{better} itself correlates with the length difference (0.06, 0.50, 0.50, 0.34), so the same pair-dependent shift may answer all three questions. With W2 (“Which response should be rejected?”) the argmax correct-reversal share is 0.000 for all four judges; W2 was not re-analyzed with the calibrated readout. RewardBench sections are unbalanced (chat 12, chat-hard 65, reasoning 223 pairs); per-section tables are in `results/analysis.md`.

A. Larger Judge as Intended Positive Control (Qwen3-4B, Post Hoc)

To check whether the negation test can register criterion following at all, we ran Qwen3-4B on resample 1 (100 pairs: 68 HEP, 32 non-HEP) with the identical prompt, readout and thinking setting. Its readout is clean (mean A/B mass 0.989; 1.0% of passes below 0.5) and its letter rates are moderate

TABLE V

ALL FIVE JUDGES RE-SCORED ON THE SAME 100 RESAMPLE-1 PAIRS (POST HOC). COLUMNS AS IN TABLES III AND IV; 95% BOOTSTRAP CIs. THE QWEN3-4B POINT ESTIMATES EQUAL RESULTS/CALIBRATED. JSON; ITS CIs HERE COME FROM A SEPARATE BOOTSTRAP DRAW, SO THEY CAN DIFFER IN THE LAST DIGIT FROM THOSE QUOTED IN THE TEXT (E.G. r UPPER BOUND 0.29 VS. 0.30).

Judge ($n = 100$)	r	Same sign	Cal. acc.	Cal. rev.	Slope	Ratio
Qwen2.5-0.5B	0.94 [0.81, 0.98]	0.93	0.65	0.05	0.75 [0.64, 0.84]	0.85
Qwen2.5-1.5B	0.66 [0.53, 0.78]	0.72	0.53	0.21	0.41 [0.30, 0.54]	0.86
Qwen3-0.6B	0.89 [0.85, 0.93]	0.87	0.63	0.08	0.65 [0.55, 0.79]	0.82
Qwen3-1.7B	0.58 [0.40, 0.73]	0.79	0.79	0.13	0.23 [0.14, 0.32]	0.35
Qwen3-4B	0.10 [−0.09, 0.29]	0.59	0.86	0.31	0.07 [−0.06, 0.20]	0.39

(P(“A”) 0.45 under *better*, 0.30 under W1). On all 100 pairs, $r(d_{\text{better}}, d_{W1}) = 0.10 [-0.09, 0.30]$, the same-sign share is 0.59 [0.49, 0.68], calibrated “better” accuracy is 0.86 and calibrated reversal is 0.31 [0.23, 0.40]. Restricting to pairs with A/B mass > 0.9 on all four passes ($n = 97$) gives $r = 0.09 [-0.11, 0.29]$. By subset, $r = 0.30 [0.10, 0.49]$ on HEP ($n = 68$, calibrated accuracy 0.96) and $-0.19 [-0.57, 0.20]$ on non-HEP pairs ($n = 32$, accuracy 0.66); the non-HEP CI is wide and includes 0. Unlike the smaller judges, its observed all-same-letter share (0.06) is below the letter-null expectation (0.17).

Because the four smaller judges were pooled over 300 pairs, Table V re-scores every judge on the same 100 resample-1 pairs. The smaller judges’ values on this subset are close to their 300-pair values. On these pairs, Qwen3-4B’s r is lower than Qwen3-1.7B’s by 0.475 (paired bootstrap, 4B minus 1.7B: $-0.475 [-0.720, -0.229]$); the same-sign share is lower by 0.20 [0.08, 0.32], calibrated reversal is higher by 0.18 [0.07, 0.29], and the OLS slope is lower by 0.158 [0.009, 0.329]; the accuracy difference, 0.07 [−0.00, 0.15], is not distinguishable from zero at this n . Fig. 3 shows both judges on the same pairs.

The 4B judge is therefore a *partial* positive control. The coupling between the two questions is much weaker, and in 0.31 of pairs the judge prefers *C* under *better* and *R* under W1 (about 36% of the 86 pairs where it prefers *C*). But a criterion-following judge would show a clearly negative r and slope; the 4B judge shows neither (slope 0.07 [−0.06, 0.20]), and its median $|d_{W1}|/|d_{\text{better}}|$ is 0.39, so the shift mostly shrinks under “worse” rather than inverting. What the comparison does show is that the readout is not insensitive by construction: on identical pairs, prompt and readout it separates Qwen3-4B from Qwen3-1.7B. Read together with the falling r within both smaller families, the data are consistent with criterion coupling weakening with size, but this rests on one family at 4B, one resample, $n = 100$, and a post-hoc analysis, so the trend is suggestive, not established. We did not compute the argmax taxonomy or the magnitude analyses of Table IV for Qwen3-4B beyond those in Table V.

VI. DISCUSSION AND LIMITATIONS

Interpretation. On this sample, for these small judges, the logit readout is dominated by a pair-dependent shift that the negated question does not invert; the question wording mostly

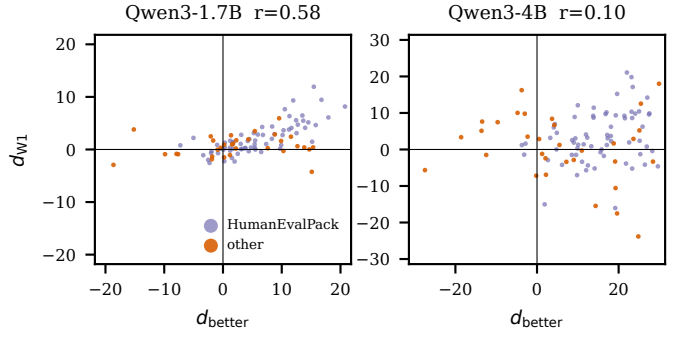


Fig. 3. Calibrated shift under “better” (x) vs. “worse” (y) for Qwen3-1.7B and Qwen3-4B on the same 100 resample-1 pairs (purple: HumanEvalPack, orange: other). A criterion-following judge would put points in the lower-right and upper-left quadrants.

moves the letter prior. For the three smaller judges the shift keeps its sign (0.71–0.92 of pairs) and most of its size (median ratio 0.82–0.90), and it is close to chance with respect to the RewardBench label. Qwen3-1.7B is different: it has content signal under “better” and its shift shrinks to about a third under “worse”, a weak partial reversal rather than none. Qwen3-4B (100 pairs) goes further: its shift is nearly uncoupled across the two questions ($r = 0.10$) but still not inverted (slope near 0). We call the small-judge pattern criterion-blind only in this operational sense. Because most pairs are one-word bug edits, it may say more about how these judges handle near-identical code than about negation in general. Our earlier, pre-registered interpretation that the smallest judges are “position-locked” and that the negation test is therefore uninformative for them was wrong: it followed from an argmax readout that a large letter offset dominates. The positive correlation may reflect any pair-level feature the judges respond to (quality, length, surface form); d cannot say which, only that it is not inverted by the negated question. For practitioners, the lesson is methodological as much as substantive: argmax verdicts of letter-biased judges can hide graded preferences, and an order-averaged logit difference is a cheap check that changes the diagnosis.

Scope. The main claim is limited to four Qwen judges up to 1.7B, a next-token argmax A/B readout (and its order-averaged logit difference), thinking disabled, one prompt template, and a 300-pair RewardBench subsample in which 217 pairs are HumanEvalPack code pairs and responses are at most 150 tokens. The Qwen3-4B result is one judge from the same family on 100 of these pairs; we make no general claim about larger judges, reasoning-enabled judges [5], or generated verdicts.

Threats to validity. (i) *Post-hoc analysis*: the calibrated readout, the letter-biased null, the subset, magnitude, size-difference, length, edit-distance and Qwen3-4B analyses were added after review; they were not pre-registered. Their CIs are descriptive, with no multiplicity control. The W2 control and the length-control shares in both orders are argmax-based. (i') *Letter offsets vary by pair*: observed same-letter shares

exceed the letter-null expectation (0.71 vs. 0.63, 0.89 vs. 0.79, 0.43 vs. 0.34), which suggests that the letter offset is not constant across pairs. If that variation depends on which text sits in slot A, it enters d (see (ii)). (ii) *Calibration model*: d removes a constant additive letter offset; order effects that interact with content are not removed. Content-by-slot interactions (e.g. an “A” logit that depends on the length of the text in slot A) enter d identically under both questions and would push r upward without any criterion-independent preference for a response. (iii) *What d measures*: d is a pair-dependent shift that the question mostly leaves unchanged, not necessarily a quality judgment. Calibrated “better” accuracy is near chance (0.50–0.61) for three judges; only Qwen3-1.7B (0.81) shows clear content signal. For Qwen2.5-0.5B, d is tiny relative to the letter offset (median $|d_{\text{better}}|$ 0.047 vs. mean offset 2.80), so its $r = 0.95$ is not evidence of a coherent preference. (iv) *Readout*: a generation check was run for Qwen3-1.7B only (first 100 pairs of resample 0, greedy, 16 tokens, regex parse); the other three judges are not validated. For Qwen3-1.7B the A/B mass is low on a quarter of passes, and the mass >0.9 subset ($n = 140$) is a selected sample. (v) *Sample*: code-heavy; HEP pairs differ by a median of one token (204 of 217 by fewer than five), so HEP results concern one-word bug edits; the non-HEP subset is small ($n = 83$); chat has 12 pairs. (vi) *Hypotheses*: H1 could not be evaluated as designed (tiny conditioning subsets); H2 is supported as pre-registered but, given the letter rates, nearly tautological. (vii) *Protocol deviation*: the development-set mass check was run for one judge only. (viii) *Novelty*: the related-work scan used arXiv and web search only.

What was not done (planned for a next cycle). (a) *Readout validation*: done for Qwen3-1.7B only (100 pairs $\times$ 4 judgments, greedy, 16 tokens; 90% agreement with logit argmax, generated reversal 4/100 vs. 1/100; results/gen_check.md); pair-level argmax categories are sensitive to the readout (category agreement 0.77; position-locked 0.22 generated vs. 0.43 logit; P(A) under “better” 0.49 vs. 0.33), so the argmax taxonomy, and in particular the position-locking share, should not be over-read; the other three judges are not validated. (b) *Positive control*: only partly done. Qwen3-4B was run on one resample (100 pairs, one family); it shows that the readout separates judges on identical pairs, but it is not a judge known to reverse (its r and slope are near 0, not negative), so the test’s sensitivity to full criterion following is still unshown. Not run: Qwen3-4B on resamples 0 and 2, another family (e.g. Qwen2.5-7B), and the argmax taxonomy and generation check for the 4B judge. (c) A non-code sample with substantively different responses was not added. (d) No thinking-enabled condition (budget rule; Section IV), no calibrated re-analysis of W2, and no prompt-wording search beyond W1/W2.

VII. CONCLUSION

Asked “which response is worse?”, four Qwen judges between 0.5B and 1.7B parameters (thinking off, A/B logit readout) almost never reverse their “better” verdict under argmax (0, 0, 0 and 1 of 300 pairs). The pre-registered diagnosis of

position-locking does not hold up: it is largely implied by the judges’ letter rates. After removing the letter offset, the pair-dependent logit shift under “worse” is positively correlated with that under “better” ($r = 0.63$ to 0.90 for the three judges that do not always answer “A”; 0.95 for Qwen2.5-0.5B, whose shift is tiny relative to its letter offset), including on non-code pairs and high-mass pairs. On this sample, which is dominated by one-word code edits, and for these small judges, the readout is dominated by a shift that the negated question does not invert; for the three smaller judges it keeps most of its size, while Qwen3-1.7B, the only judge with clear content signal (calibrated accuracy 0.81), shows an attenuated shift (weak partial reversal or lower confidence under “worse”). The correlation falls with size within both families, a post-hoc observation on two sizes each. A Qwen3-4B judge on 100 of the pairs shows a much weaker coupling ($r = 0.10$, 0.475 below Qwen3-1.7B on the same pairs) and calibrated reversal in 0.31 of pairs, but no negative correlation; with one family, one resample and $n = 100$, a scale trend is suggestive, not established. We ran a generation check for Qwen3-1.7B only (90% agreement with the logit argmax, 100 pairs).

REPRODUCIBILITY STATEMENT

Repository path: projects/small-judge-reversal/. Raw per-pass outputs: results/raw_<model>_r<k>.json; model and data revisions: results/models.md; environment: results/pip_freeze.txt; commands: results/commands.md. From the project directory:

```
uv venv .venv --python 3.14
uv pip install --python .venv/bin/python \
  -r requirements.txt
./fetch_data.sh
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/run.py prepare
# for M in the 4 models, s in 0 1 2:
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/run.py run --model $M --resample $s
# Qwen3-4B positive control (resample 1 only):
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/run.py run --model Qwen3-4B --resample 1
.venv/bin/python -I src/analyze.py
# generation check (seed 0; see results/gen_check*):
OMP_NUM_THREADS=4 .venv/bin/python -I \
  src/gen_check.py gen --model Qwen3-1.7B --n 100
.venv/bin/python -I src/gen_check.py report
.venv/bin/python -I src/calibrated.py
.venv/bin/python -I src/referee2.py
# venv with paper/requirements-paper.txt:
python -I paper/make_figures.py
```

Only src/analyze.py, src/calibrated.py, src/referee2.py and paper/make_figures.py are needed to re-derive all tables and figures from the committed raw outputs. Caveats for the Qwen3-4B run: fetch_data.sh does not download Qwen3-4B (fetch the pinned revision separately), and because the model config now lists Qwen3-4B, for which only resample 1 exists, src/analyze.py must skip that model to run.

AI-AUTHORSHIP DISCLOSURE

Produced by an autonomous AI agent (Claude) on behalf of @pranavndyalam. Not peer reviewed. The study design, code, experiments, analysis and this text were produced by the

agent; the plan, results and a first version of this paper were reviewed by automated overseer, ethics and referee agents, not by human peer review.

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